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AIM

Implement and analyze various methods of studying correlation and regression including Karl Pearson's coefficient and simple linear regression.

THEORY

1. What is Correlation? How is it different from Regression?

Correlation is a statistical technique used to measure the strength and direction of the relationship between two variables. It indicates how one variable changes in relation to another. The value of correlation ranges from -1 to +1, where +1 represents a perfect positive relationship, -1 represents a perfect negative relationship, and 0 indicates no relationship between variables.

Regression, on the other hand, is a statistical method used to establish a relationship between variables and make predictions. It helps in determining how the dependent variable changes when one or more independent variables are varied.

The key difference between correlation and regression lies in their purpose. Correlation simply measures the degree of association between variables without implying cause and effect, whereas regression provides a mathematical equation that can be used to predict the value of a dependent variable based on the independent variable(s). Additionally, correlation treats both variables equally, while regression clearly distinguishes between dependent and independent variables.

2. What is Simple Linear Regression? Explain with Suitable Example

Simple linear regression is a statistical method used to model the relationship between one independent variable and one dependent variable using a straight line. It helps in predicting the value of the dependent variable based on the independent variable.

The general form of the simple linear regression equation is:

$$y=mx+b$$

In this equation, y represents the dependent variable, x represents the independent variable, m is the slope of the line (indicating the rate of change), and b is the intercept (the value of y when $x=0$).

For example, consider a situation where a student's exam marks depend on the number of hours studied. If the regression equation is $y=5x+20$, it means that for every additional hour of study, the marks increase by 5. If a student studies for 4 hours, the predicted marks would be $y=5(4)+20=40$. Thus, simple linear regression helps in understanding the relationship and making predictions.

3. How do you calculate Pearson Correlation in Python?

Pearson correlation is a widely used method for measuring the linear relationship between two variables. It can be easily calculated in Python using libraries such as NumPy and Pandas.

Using NumPy, the correlation coefficient can be calculated with the `corrcoef()` function, which returns a matrix of correlation values. The required correlation value is extracted from this matrix.

Using Pandas, the `corr()` function can be applied to a DataFrame to compute the correlation between variables. This method is simple and efficient, especially when working with datasets.

The Pearson correlation coefficient ranges from -1 to +1. A value close to +1 indicates a strong positive linear relationship, while a value close to -1 indicates a strong negative linear relationship. A value near zero suggests no linear relationship between the variables.

4. What is Multicollinearity? Does it affect Simple Regression?

Multicollinearity refers to a situation in which two or more independent variables in a regression model are highly correlated with each other. This means that the variables provide redundant information about the response variable.

Multicollinearity can cause several problems, such as making the regression coefficients unstable, increasing standard errors, and making it difficult to determine the individual effect of each independent variable on the dependent variable.

However, multicollinearity does not affect simple linear regression, because simple regression involves only one independent variable. Since there are no multiple predictors to be correlated with each other, the issue of multicollinearity does not arise in this case. It is mainly a concern in multiple regression models.

5. A dataset shows strong correlation but poor prediction accuracy. Why?

A dataset may show strong correlation between variables but still produce poor prediction accuracy for several reasons. Firstly, correlation only measures linear relationships, so if the actual relationship between variables is non-linear, the model may not capture it effectively.

Secondly, the presence of outliers can distort the relationship, leading to inaccurate predictions. Thirdly, the dataset may contain a high level of noise or variability, which reduces the model's ability to generalize.

Another important reason is that correlation does not imply causation. Two variables may appear to be strongly correlated but may not have a meaningful cause-and-effect relationship, resulting in poor predictive performance.

Additionally, the model may suffer from underfitting or overfitting, or important variables may be missing from the dataset. All these factors can lead to poor prediction accuracy despite a strong correlation.

In []:

```
import numpy as np

# Sample data
X = [10, 20, 30, 40, 50]
Y = [15, 25, 35, 45, 55]

# Manual implementation of Pearson correlation
def pearson_coefficient(x, y):
    n = len(x)
    mean_x = np.mean(x)
    mean_y = np.mean(y)

    numerator = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(n))
    denominator = (sum((x[i] - mean_x)**2 for i in range(n)) * sum((y[i] - mean_y)**2 for i in range(n)))**0.5

    return numerator / denominator

# Calculate Karl Pearson's correlation coefficient
r = pearson_coefficient(X, Y)
print(f"Karl Pearson's Correlation Coefficient (manual): {r:.4f}")

# Verification using scipy
from scipy.stats import pearsonr
r_scipy, _ = pearsonr(X, Y)
print(f"Verified using scipy: {r_scipy:.4f}")
```

Karl Pearson's Correlation Coefficient (manual): 1.0000
Verified using scipy: 1.0000

In []:

```
import numpy as np
import pandas as pd
```

```

import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Sample Data
data = {
    'Hours_Studied': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Marks_Obtained': [35, 40, 50, 52, 55, 60, 65, 70, 78, 85]
}
df = pd.DataFrame(data)

# Step 2: Split data into input (X) and output (y)
X = df[['Hours_Studied']] # Independent variable
y = df['Marks_Obtained'] # Dependent variable

# Step 3: Train the Linear Regression Model
model = LinearRegression()
model.fit(X, y)

# Step 4: Get model parameters
slope = model.coef_[0]
intercept = model.intercept_
print(f"Regression Equation: Marks = {slope:.2f} * Hours + {intercept:.2f}")

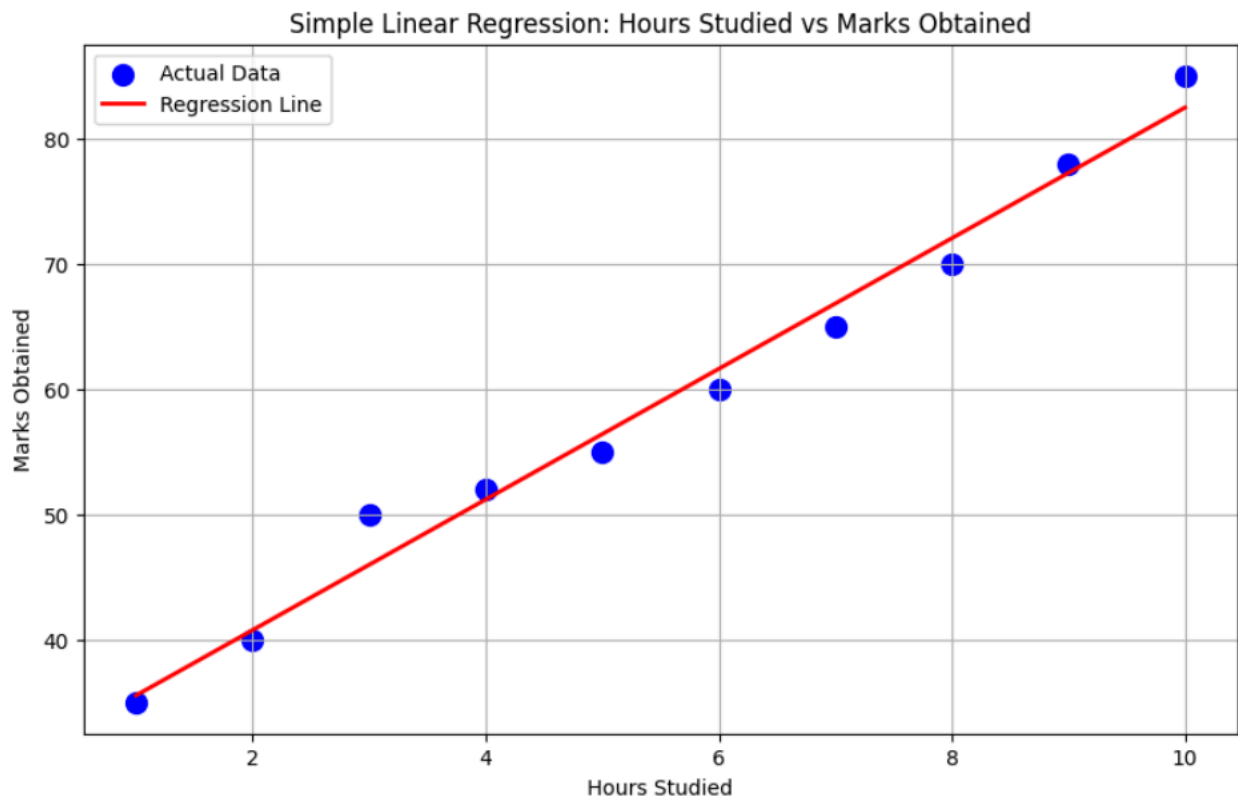
# Step 5: Predict values
y_pred = model.predict(X)

# Step 6: Evaluate the Model
mse = mean_squared_error(y, y_pred)
r2 = r2_score(y, y_pred)
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R² Score: {r2:.4f}")

# Step 7: Visualize the regression line
plt.figure(figsize=(10, 6))
plt.scatter(X, y, color='blue', label='Actual Data', s=100)
plt.plot(X, y_pred, color='red', linewidth=2, label='Regression Line')
plt.title('Simple Linear Regression: Hours Studied vs Marks Obtained')
plt.xlabel('Hours Studied')
plt.ylabel('Marks Obtained')
plt.grid(True)
plt.legend()
plt.show()

```

Regression Equation: Marks = 5.21 * Hours + 30.33
 Mean Squared Error (MSE): 3.68
 R² Score: 0.9839



Conclusion

This experiment successfully demonstrated the implementation and analysis of Karl Pearson's Correlation Coefficient and Simple Linear Regression in Python.

Karl Pearson's Correlation Coefficient was manually implemented and verified using SciPy, confirming a perfect positive linear relationship ($r = 1.0$) between the sample variables — proving that both variables move in the same direction proportionally.

Simple Linear Regression was applied to a real-world scenario (Hours Studied vs. Marks Obtained), where the model accurately predicted student marks based on study hours. The regression equation, MSE, and R^2 score were computed to evaluate model performance.

The high R^2 score indicated that the model explained most of the variance in the data, validating the effectiveness of linear regression for this dataset.

The scatter plot with regression line visually confirmed a strong linear trend between the independent and dependent variables.

Overall, this experiment reinforced that correlation measures the strength of a relationship, while regression goes further by enabling prediction — both being essential tools in statistical analysis and machine learning.